ROLLS THEOREM , MEAN VALUE THEOREM

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ROLLS THEOREM

If $f(x)$ is defined and continuous on $a \le x \le b$

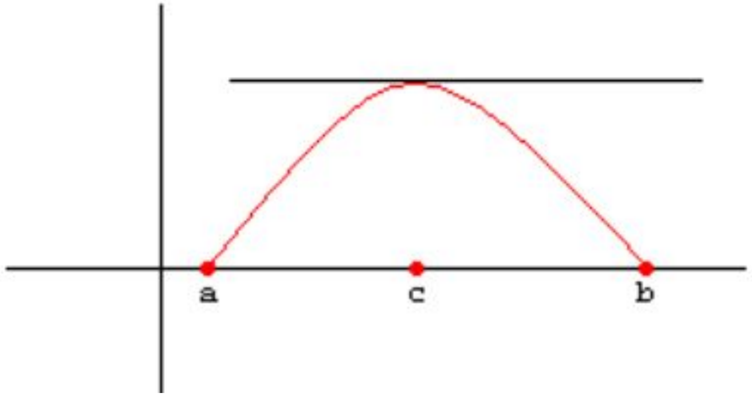
and differentiable on $a < x < b$

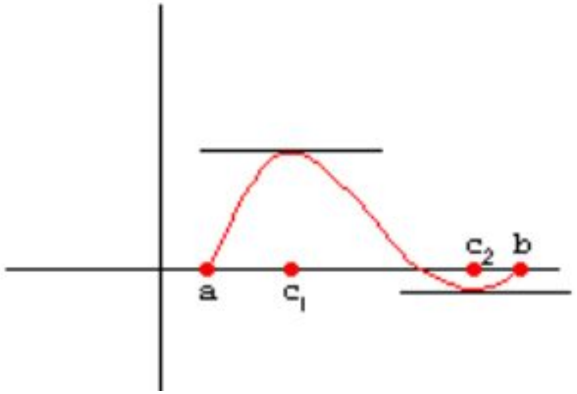
and $f(a) = f(b) = 0$,

Then there is at least one number c between a and b

where $f'(c) = 0$.

(That is, $f'(c) = 0$ for some c, $a < c < b$)





3. Example of Rolle's Theorem

Given $y = x^3 - 4x$

Find a value in the interval $(-2, 2)$ which satisfies Rolle's Theorem.

First, verify that Rolle's Theorem can be applied:

$f(-2) = 0$ and $f(2) = 0$ and $f(x)$ is continuous and differentiable

Then $f'(x) = 3x^2 - 4$

There must be at least one value of x between -2 and 2 for which $f'(x) = 0$.

Set $3x^2 - 4 = 0$

$$x^2 = \frac{4}{3}$$

$x = \pm\dfrac{2\sqrt{3}}{3}$, so there are two values which satisfy Rolle's Theorem!

MEAN VALUE THEOREM

Theorem:

If $f(x)$ is defined and continuous on $a \le x \le b$

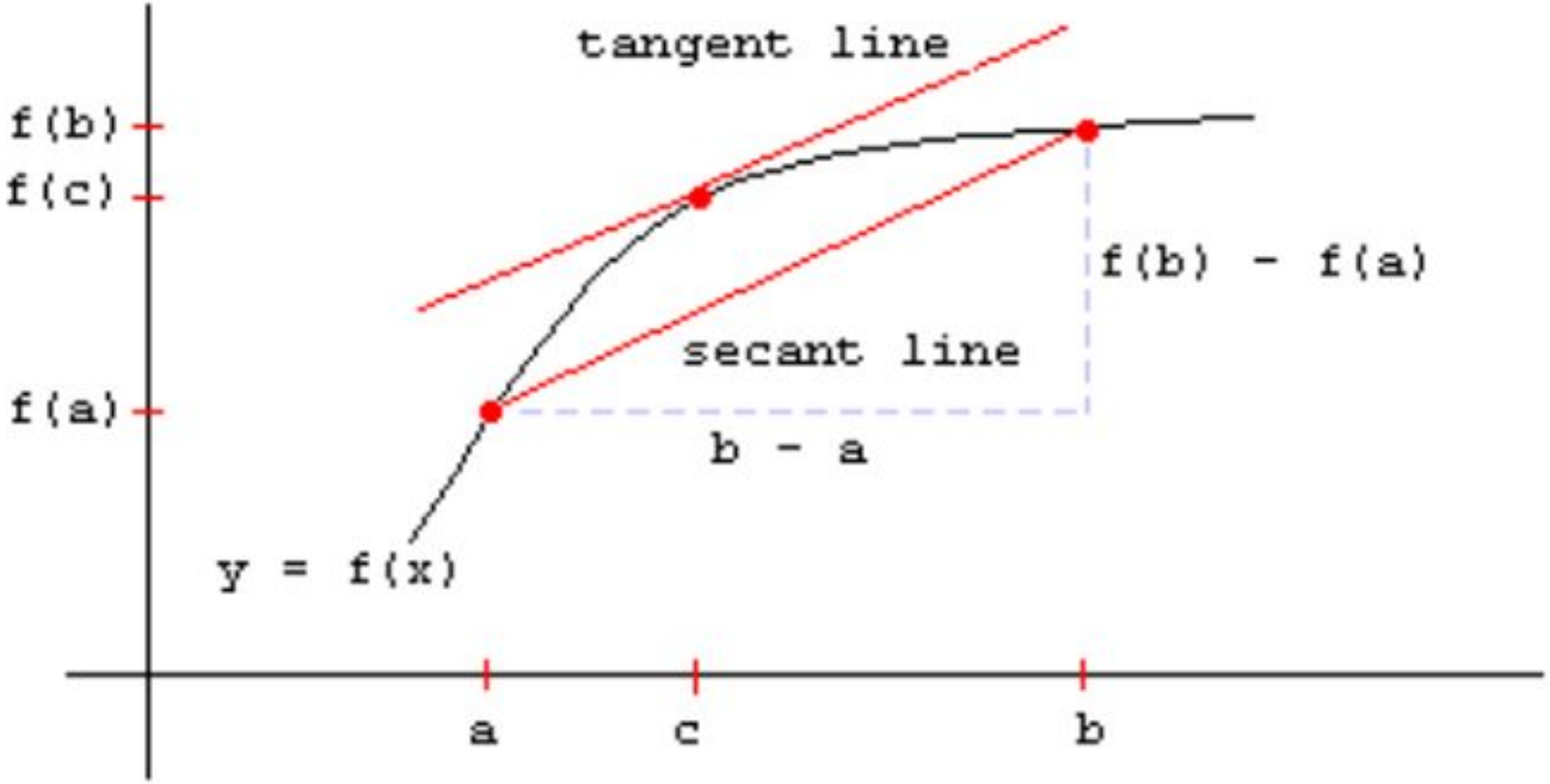
and differentiable on $a < x < b$,

Then there is at least one number c between a and b

where $f(b)-f(a)=f'(c)(b-a)$

In other words, $f'(c)=\dfrac{f(b)-f(a)}{b-a}$

Geometrically, there must be at least one point in the interval where the tangent to the curve at that point is parallel to the secant line which passes through the points (a, f(a)) and (b, f(b)).



tangent line
f(b)
f(c)
f(b) - f(a)
secant line
f(a)
b - a
y = f(x)
a
c
b

Given $f(x) = x^3$ and a = -2 and b = 2, find the points which satisfy the Mean Value Theorem.

Solution: $f'(x) = 3x^2$

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$3c^2 = \frac{8 - (-8)}{2 - (-2)}$$

$$3c^2 = \frac{16}{4}$$

$$3c^2 = 4$$

$$c^2 = \frac{4}{3}$$

$$c = \pm\frac{2\sqrt{3}}{3}$$